PIC Process Flows (as implemented)

AlphaHelion Purchasing Intelligence Cloud · AHG/PRC/PIC/2026-10/06 · 27 October 2026

This document describes the nine business flows F1–F9 as they now run in the backend (`backend/app/pic`), on PostgreSQL (Neon) in schema `pic` . Each flow lists:

- the swimlane diagram (Blueprint §21 steps, same step IDs),
- the API call that moves each step,
- the rules, SoD checks and guards enforced at each gate,
- the token path recorded in `process_step` for the Axion Mobility signature case.

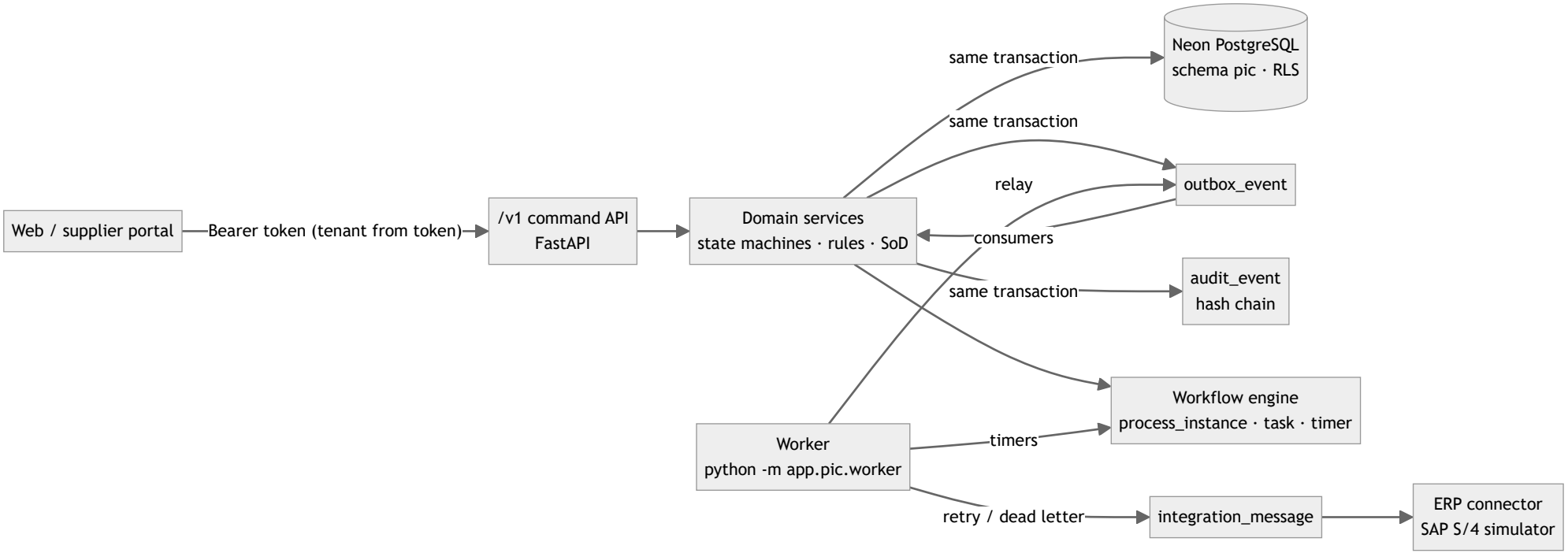
Every step a document passes is stored, so any document's live position can be read with `GET /v1/documents/{TYPE}/{ref}/process` .

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0. How the flows run



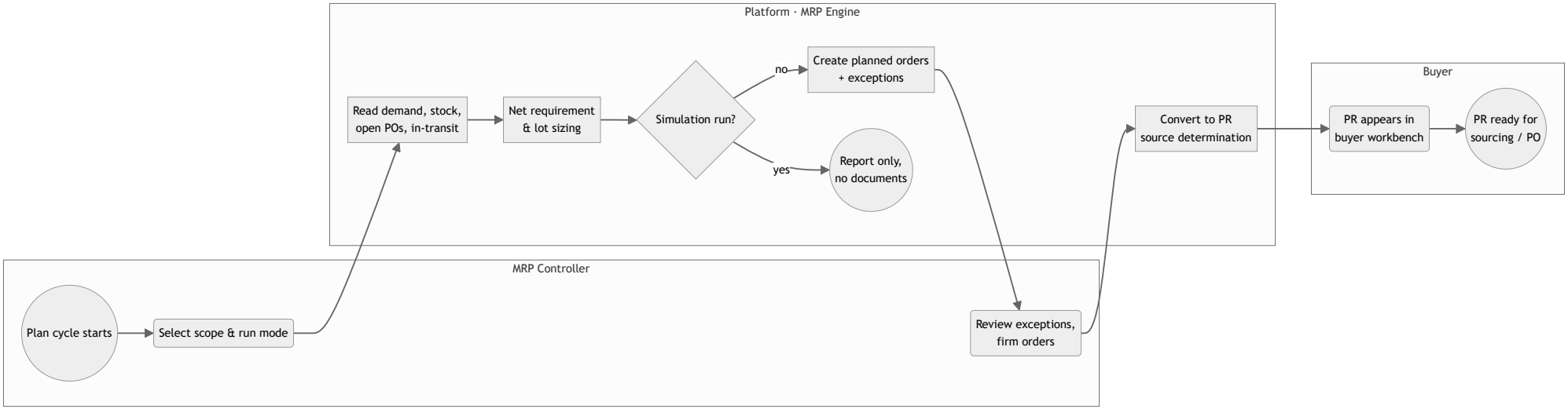
Element	Implementation
Tenant isolation	Every request runs `SET LOCAL ROLE pic_app` and `app.tenant_id`; 85 tables have forced row-level security.
Workflow engine	Postgres-native engine standing in for Camunda 8: process instances, step history, human tasks (sequential or parallel), timers.
Rules	Effective-dated `rule_set` rows (DMN-01, DMN-02, DMN-03, PO-RISK, …). Every document stores the versions it used.
SoD	`sod_rule` + `action_log`, checked at command time for people and agents. Blocked attempts are written to `sod_violation`.
Events	Transactional outbox; consumers are idempotent (`processed_event`); failures retry up to 8 times.
ERP	`integration_message` with a unique idempotency key. The ERP is queried before any retry; failures dead-letter, and can be replayed from `/v1/integration/messages/{id}/replay`.

Element	Implementation
Audit	Append-only (the `pic_app` role cannot UPDATE or DELETE it). SHA-256 hash chain per tenant, verified at `GET /v1/audit/verify`.
Errors	RFC 9457 `application/problem+json` with `code`, `ruleRef` and `traceId`. Illegal transitions return 409; stale writes return 412.

Legend used in the diagrams: rounded box = user task, square box = system task, diamond = gateway, circle = start/end event.

F1 — MRP run → planned order → purchase requisition

Process key `pic.mrp-to-pr` · Module M02 · Service `services/planning.py`



Step	API	What the code enforces
F1-02	`POST /v1/mrp/runs {plant, mode}`	Modes `NET_CHANGE` (materials whose demand changed since the last run), `REGENERATIVE`, `SIMULATION`. Only materials with MRP type `PD`.
F1-03/04	(inside the run)	Daily buckets. Projected stock = on-hand + open PO + in-transit + firm planned receipts − gross demand. A planned order is created when projected stock drops below safety stock. Lot sizing: exact, fixed, EOQ, period, min/max, pack rounding.
F1-04		Release date = need date − (planned delivery + GR processing days) on the plant's factory calendar.
F1-05		Simulation stores the netting trace in `mrp_run.results` and ends without documents (tested).
F1-06		Unfirmed planned orders from earlier runs are deleted; firmed ones are never changed (RESCHEDULE_IN/OUT exceptions instead). Exceptions: PAST_DUE, BELOW_SAFETY, RESCHEDULE_IN.
F1-07	`POST /v1/planned-orders/{id}/firm`	Event `planned_order.firmed`.
F1-08	`POST /v1/planned-orders/{id}/convert`	Creates a PR with origin `MRP` (P02, immutable by database trigger), inheriting material, quantity, date, plant and storage location. Source determination order: contract → quota → source list → info record. The PR then goes through the same F2 approval controls.

Step	API	What the code enforces
F1-09/10	(on PR approval)	The run instance ends when every planned order is converted.

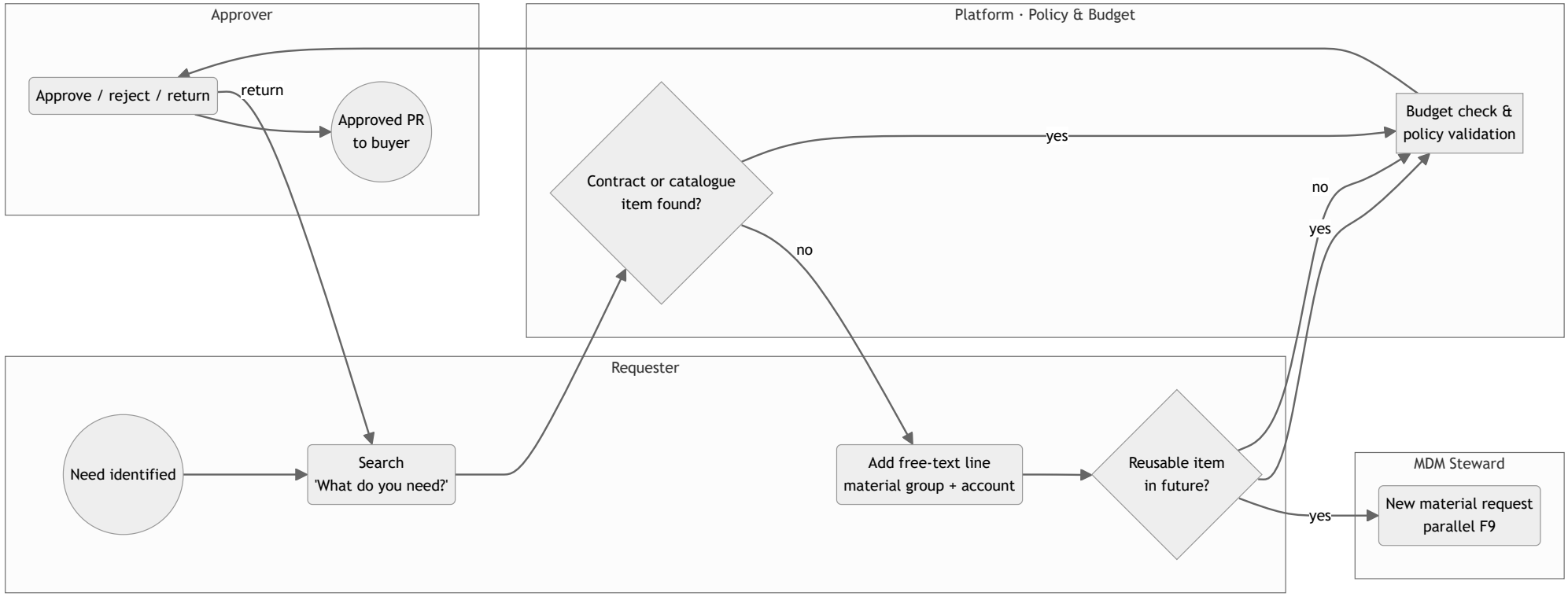
Recorded path, MR-1043:

Step	Outcome
F1-02	plant P100, net-change
F1-03	2 materials in scope
F1-04	DC-330: 12,000 − (1,400 + 1,200 + 600 + 0) + 200 = **9,000**
F1-06	1 planned order (PLO-55821, need 12 Nov, release 30 Oct), 1 exception (DC-331 BELOW_SAFETY)
F1-07	PLO-55821 firmed (mrp.controller)
F1-08	REQ-2311 created; source: info record
F1-09/10	REQ-2311 in buyer workbench after the 4-step approval

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F2 — Guided buying → requisition

Process key `pic.requisition` · Module M04 · Service `services/requisition.py`



Step	API	What the code enforces
F2-02	`GET /v1/guided-buying/search?q=`	**P03** search order: contract item → catalogue item → material master → free text. The response says which channel to use and whether free text is allowed.
F2-02/04	`POST /v1/requisitions`	**P04** free-text lines need a material group and can never reference a material (DB `CHECK free_text_no_material`). Purchase type (P08) per line.
F2-05/06	(on create)	Lines flagged `reusable` start an F9 material request in parallel; the PR does not wait.

Step	API	What the code enforces
F2-07	`POST /v1/requisitions/{id}/submit`	**P01** purpose code mandatory. Non-stock/service/expense/asset lines need an account assignment. **Quote policy:** free text below ₹25,000 needs none, below ₹2 L one, above that three quotes or a sourcing event. **Budget:** hard stop for asset/WBS, soft stop with justification for opex; commitment booked. **DMN-01** approver chain.
F2-08	`GET /v1/tasks`, `POST /v1/tasks/{id}/complete`	Sequential approval. The requester and earlier approvers are excluded (four-eyes). Cost-object owner and department head are matched by ABAC (`cost_objects`, `plants`). SLA 2 business days, then escalation timer. Reject and return need a comment; reject/return release the budget commitment.
after	`PATCH /v1/requisitions/{id}/lines/{n}`	A change after approval restarts approval only if value, account or supplier changes.

DMN-01 approval route (`rule_set` DMN-01@2026.1, values in ₹):

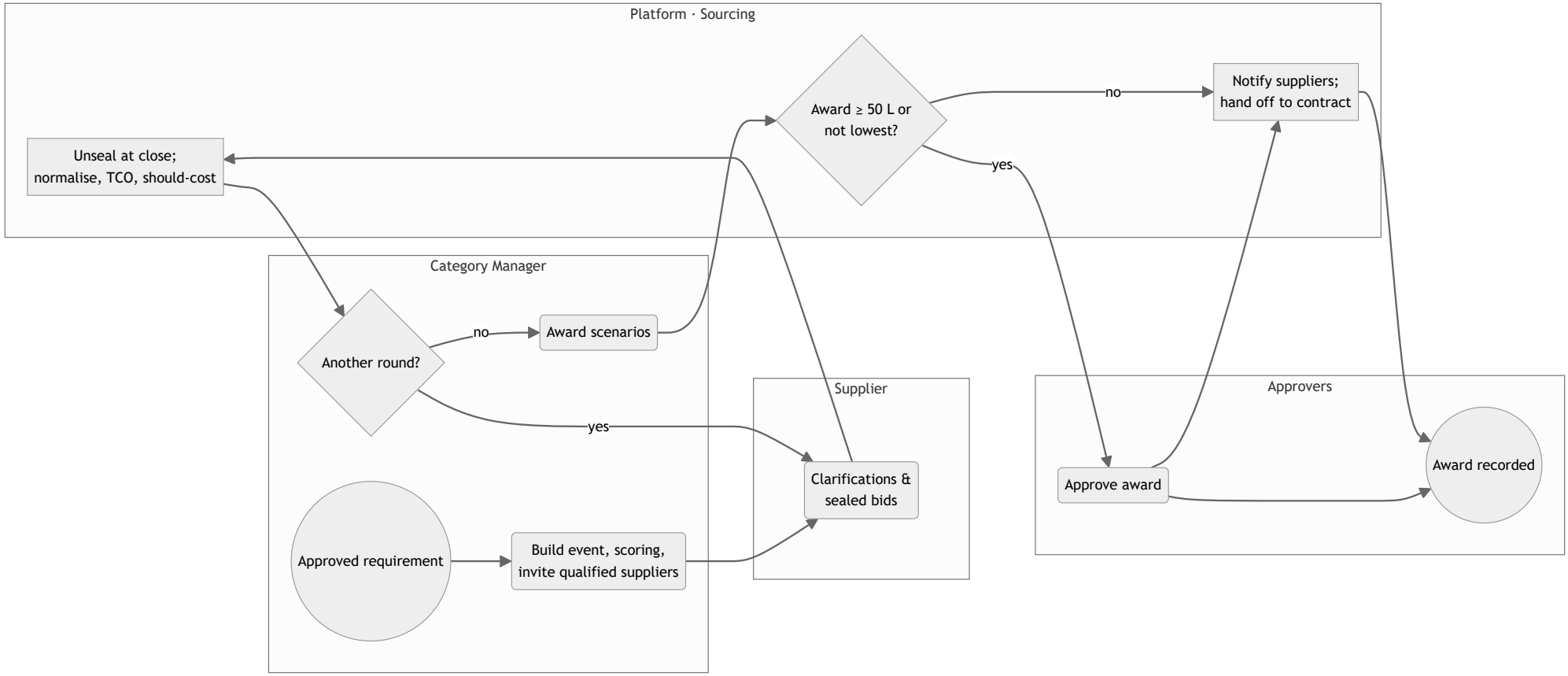
PR value	Approvers
< 25,000, catalogue or contract	auto-approved
< 2,00,000	cost-object owner
2 L – 10 L	cost-object owner → department head
> 10 L	cost-object owner → department head → procurement head → finance controller
asset	+ capex committee

Recorded paths:

- **REQ-2311** (origin MRP, ₹36.9 L):
  - F2-07 budget not applicable (inventory purchase); route of 4 approvers.
  - F2-08 approved by COST_OBJECT_OWNER → DEPT_HEAD → PROCUREMENT_HEAD → FINANCE_CONTROLLER.
  - F2-09 to buyer workbench.
- **REQ-2312** (free-text fixture, ₹1.85 L, CC 4100):
  - F2-04 free text, F2-05 reusable → F2-06 MRQ-7731 started.
  - F2-07 budget OK, route COST_OBJECT_OWNER.
  - F2-08 approved by plant.controller.

F3 — Strategic sourcing → award

Process key `pic.sourcing-award` · Module M05 · Service `services/sourcing.py`



Step	API	What the code enforces
F3-02	`POST /v1/sourcing/events`	**P06**: lines only from approved requisitions; invitees only if Qualified for the category (Conditional needs a written waiver). Should-cost model; baseline by method (last price paid from the info record).
F3-02	`POST /v1/sourcing/events/{id}/publish`	Weights must sum to 100 and lock at publish. Close timer created. `sourcing.event.published` → savings initiative *Identified*.
F3-03	`POST /v1/sourcing/events/{id}/bids` (supplier)	Bids encrypted with a per-tenant key, SHA-256 hash recorded. Reading bids before close → **423 BIDS_SEALED** and an audited `BID_ACCESS_DENIED`.

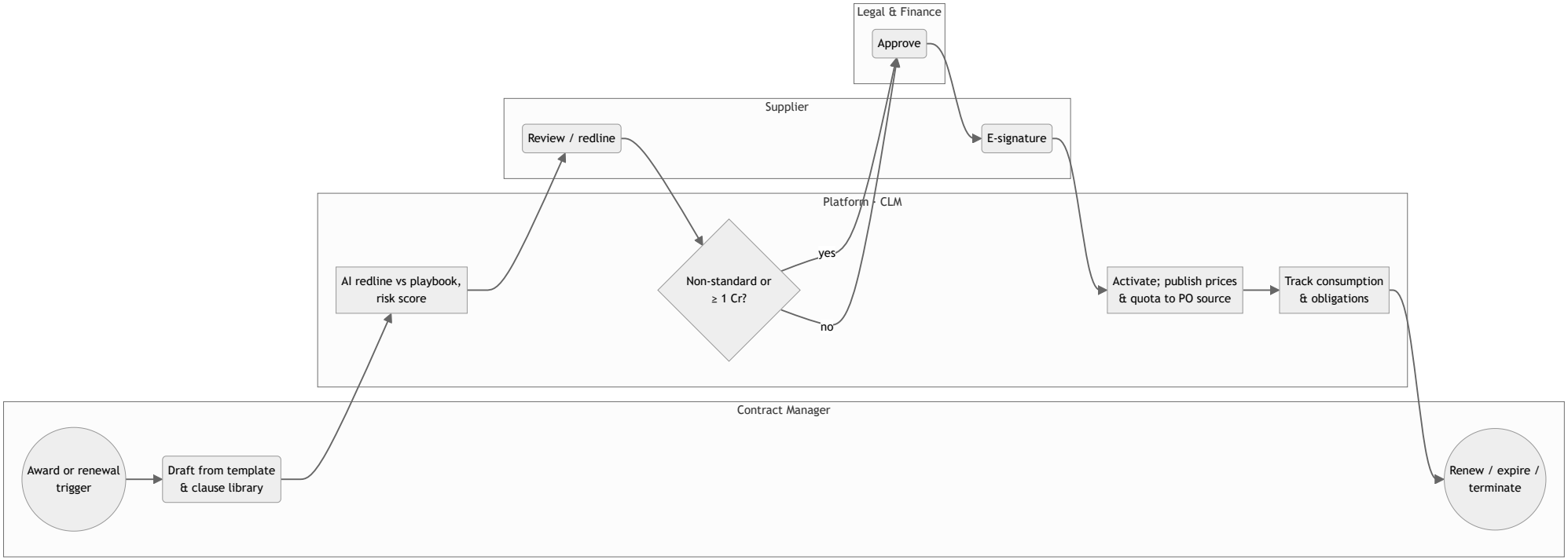
Step	API	What the code enforces
F3-03/04	close timer or `POST …/close`	At close: unseal, verify hash, compute TCO = price + freight + duty + packaging + carrying(lead time) + quality(PPM × CoPQ) − payment-term value. Bid Analysis agent run (L2).
F3-04	`POST …/scores`	Manual criteria per evaluator; PRICE_TCO is computed. Each scoring action is logged for **SOD-05**.
F3-05	`POST …/rounds`	Negotiation round; previous-round bids are superseded per supplier.
F3-06	`POST …/scenarios`	LOWEST_COST, BEST_VALUE and CAPACITY_CONSTRAINED. The optimiser minimises Σ x·TCO subject to Σx = 1, x·Q ≤ verified twin capacity, min ≤ x ≤ max share, and the maximum number of suppliers (exact enumeration).
F3-06/07	`POST …/award`	Non-lowest award needs a rationale; conflict-of-interest declaration mandatory; **SOD-05** (the sole evaluator cannot award).
F3-08	tasks	AWARD-APPROVAL: below ₹50 L category manager; ₹50 L and above category manager → CPO → finance controller. The proposer is excluded.
F3-09		`sourcing.event.awarded` → contract drafts (F4) and savings *Negotiated*.

Recorded path, RFQ-2231:

Step	Outcome
F3-02	published; weights locked
F3-03	round 1 closed by timer (agent:SCHEDULER); 3 bids unsealed
F3-04	TCO: TF001 398.00→402.50 · KD002 412.00→414.94 · SLA003 405.00→421.80
F3-05	yes: negotiation round 2
F3-03/04	round 2: KD002 405.00→408.01
F3-06	CAPACITY_CONSTRAINED proposed (LOWEST_COST and BEST_VALUE infeasible: capacity 60k/45k/70k < 90k/month)
F3-07	yes: annual value ₹43,28,64,000.00, not the lowest bid
F3-08	approved by CATEGORY_MANAGER → CPO → FINANCE_CONTROLLER
F3-10	TF001 60% @ 398.00 / KD002 40% @ 405.00; **weighted 400.80**

F4 — Contract lifecycle

Process key `pic.contract` · Module M07 · Service `services/contract.py`



Step	API	What the code enforces
F4-01/02	consumer of `sourcing.event.awarded`	One scheduling agreement per awarded supplier: target quantity = share × annual quantity; price validity from the award; tolerances +2/−5; LME index ±3% band; clauses and obligations from the CLM playbook.
F4-03	`POST /v1/contracts/{id}/review`	Contract Review agent (L2): clause positions STANDARD 0 / FALLBACK 30 / NON_STANDARD 70, with fallback suggestions.
F4-04	`POST /v1/contracts/{id}/redlines`	Supplier redlines are audited with before and after text.
F4-05/06	`POST /v1/contracts/{id}/submit`	₹1 Cr and above, or any non-standard clause → LEGAL → FINANCE_CONTROLLER → CPO; otherwise contract manager.
F4-07	`POST /v1/contracts/{id}/signatures`	Buyer (`contract:sign`) and supplier (portal user) both sign the same document hash.

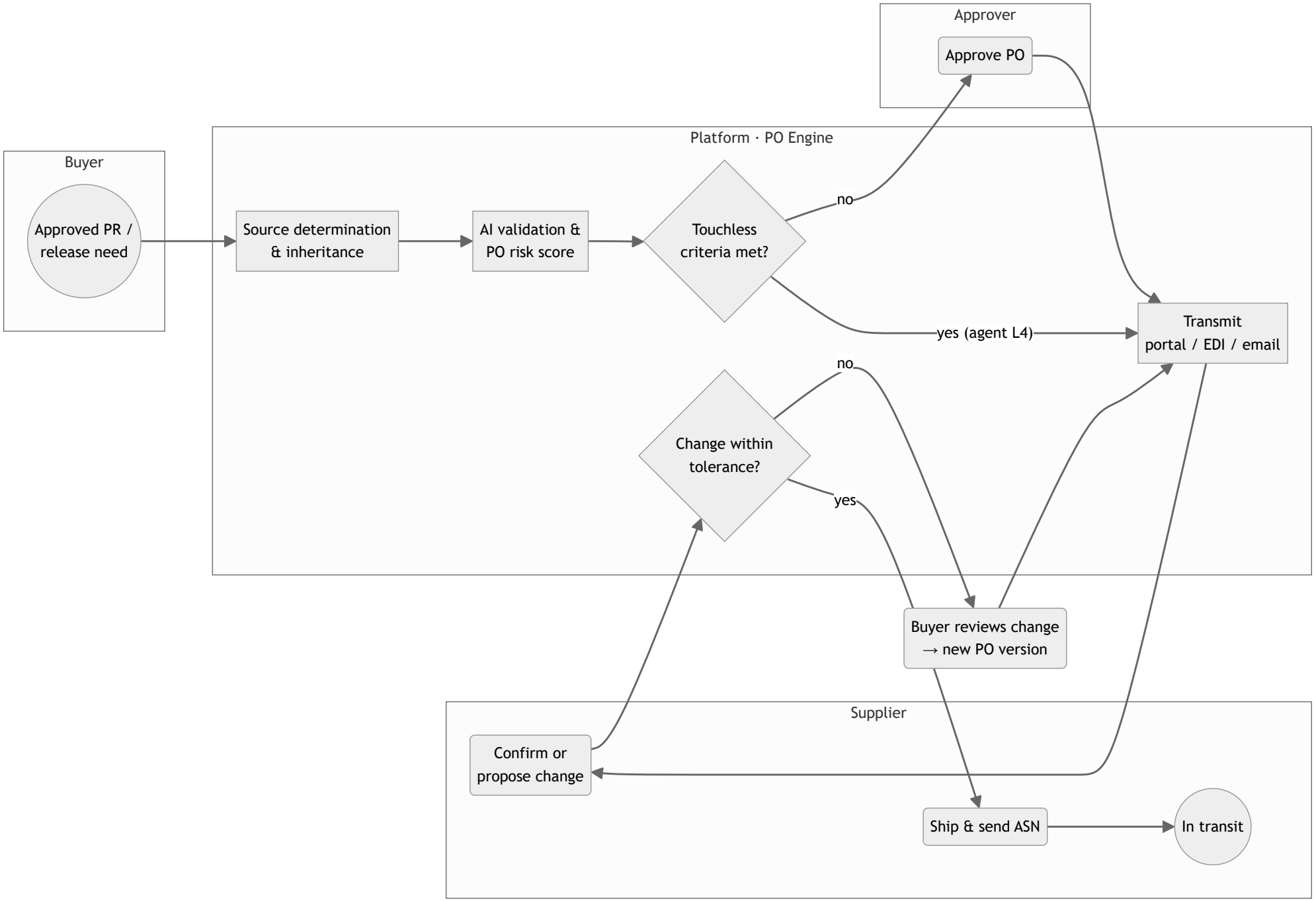
Step	API	What the code enforces
F4-08	`POST /v1/contracts/{id}/activate`	Requires all signatures with matching hashes. Publishes price validity to source determination and creates the 60/40 quota arrangement. `contract.activated` → savings *Contracted*. Renewal timer at −120 days; expiry timer.
F4-09	`POST /v1/contracts/{id}/index-resets`, `GET …/consumption`	Index resets always add a new validity period and never overwrite. Within the band the price is unchanged and the absorbed increase is booked as **cost avoidance**. Consumption alerts at 80/90/100% (`contract.consumption.threshold`). Off-contract leakage is listed.

Recorded path, SA-5520:

Step	Outcome
F4-02	drafted by the event relay with 6 clauses
F4-03	contract risk 0, no deviations
F4-05	yes: value ₹25,79,04,000.00
F4-06	approved by LEGAL → FINANCE_CONTROLLER → CPO
F4-07	signed by buyer and supplier
F4-08	active; price published
F4-09	tracking; reset on 1 Jan 2027 at +0.74% absorbed

F5 — Purchase order → confirmation → delivery

Process key `pic.po` · Module M08 · Service `services/order.py`


Approver
Approve PO
Platform · PO Engine
Buyer
Approved PR / release need
Source determination & inheritance
AI validation & PO risk score
Touchless criteria met?
no
yes (agent L4)
Transmit portal / EDI / email
no
Change within tolerance?
yes
Buyer reviews change → new PO version
Supplier
Confirm or propose change
Ship & send ASN
In transit

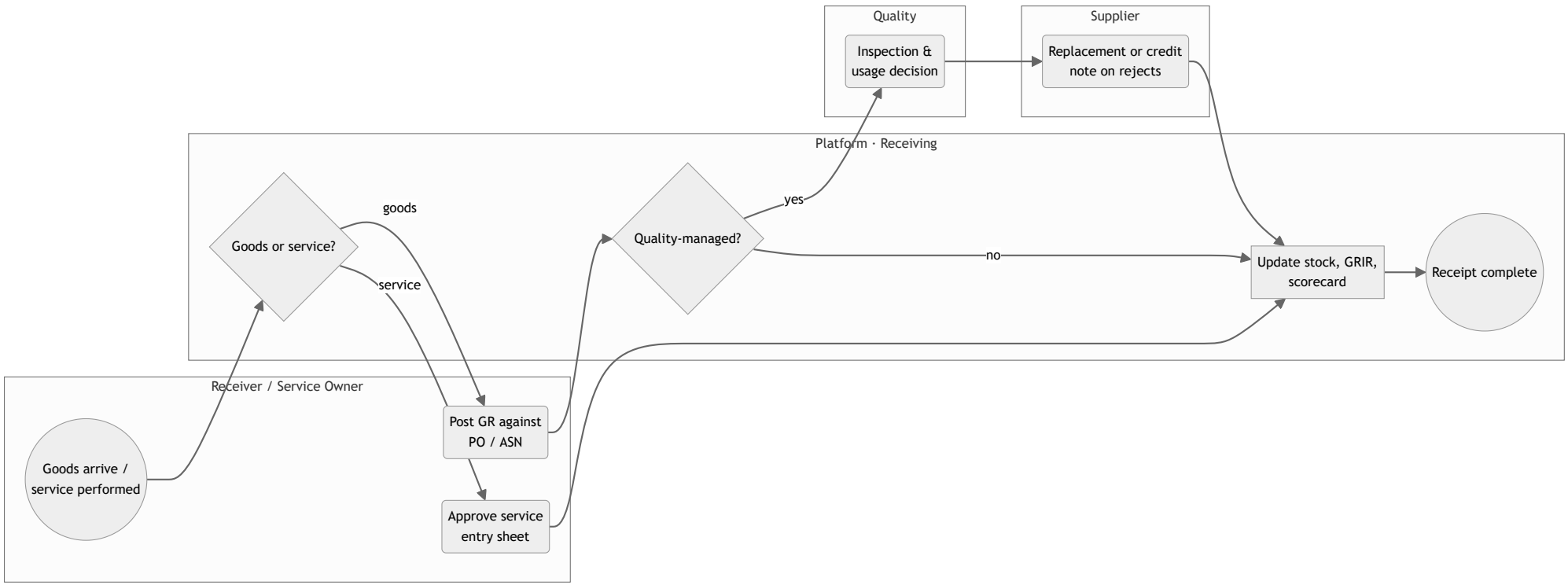
Step	API	What the code enforces
F5-02	`POST /v1/purchase-orders {source:{type:REQUISITION,id,lines}, supplierId?, contractRef?, release?}`	**P07** inheritance of PR, contract price valid on the need date, payment terms, Incoterms and tolerances. A quota splits the requirement (REQ-2311 9,000 → 5,400 + 3,600). A manual price outside contract tolerance needs a `deviationReason` (DB CHECK). **P08** purchase-type matrix sets GR/IR/SES/ERS flags. **SOD-01**: the supplier-master creator cannot create the PO.
F5-03	`POST /v1/purchase-orders/{id}/validate`	PO Validation agent (L3). Additive risk: 18 if late-delivery probability ≥ 0.15, +35 if ≥ 0.30, 18 for price deviation, 40 if the supplier is not qualified, 15 if the value is above the band, 25 for a split pattern, 12 if the account assignment changed.
F5-04		**DMN-02** touchless: contract/catalogue price, supplier Qualified/Preferred, value ≤ ₹25 L, no split, late < 0.30, risk < 30. The Touchless PO agent (L4) releases only after its autonomy guard passes (kill switch, autonomy level, scope, value limit, acceptance rate).
F5-05	tasks	Otherwise PO-APPROVAL: below ₹5 L a second buyer, above that also the procurement head. The creator is excluded. A blocked supplier → **422 PO_SUPPLIER_BLOCKED**.
F5-06		Status RELEASED → TRANSMITTED; confirmation SLA 24 h (timer → Expediting agent draft and buyer task). `purchase_order.released` → ERP PO posting and contract consumption.
F5-07/08	`POST /v1/purchase-orders/{id}/confirmations` (supplier)	Hours to confirm recorded. Changes within tolerance are accepted; changes outside go to a buyer task.
change	`POST /v1/purchase-orders/{id}/versions` (If-Match)	Version n+1 snapshot in `po_version`; re-confirmation required; 412 on a stale ETag.
F5-09/10	`POST /v1/asns`	Batches, e-way bill and CoC documents; ship quantity ≤ ordered + over-tolerance. Service POs hand over to F6 service entry instead.

Recorded paths:

- **PO-77410:**
  - F5-02 TF001, inherited from REQ-2311 and SA-5520.
  - F5-03 risk 18/100 (LATE_RISK_MED +18).
  - F5-04 yes. F5-05 skipped, released by agent:TOUCHLESS_PO (L4).
  - F5-06 via portal. F5-07 confirmed in 6.00 h. F5-09 ASN-61208 with 2 batches. F5-10 in transit.
- **PO-77412** (service, manual price):
  - F5-04 no: price source MANUAL.
  - F5-05 approved by BUYER (buyer.lead).
  - F5-07 confirmed. F5-09 handed over to service entry.

F6 — Goods receipt / service entry → quality

Process key `pic.receipt` · Module M09 · Service `services/receiving.py`



Step	API	What the code enforces
F6-03	`POST /v1/goods-receipts`	**SOD-03**: the PO creator cannot post the GR. Over-delivery beyond tolerance is blocked. Service lines are refused (they need an SES). Movement 101 to QUALITY or UNRESTRICTED stock. Batch, serial and expiry captured. ASN received quantities updated.
F6-05/06	`POST /v1/inspection-lots/{id}/usage-decision`	Sample size per ISO 2859-1 (5,400 → 200). Accepted + rejected must equal the lot; rejections need a defect code and reason. Movement 321 to unrestricted, 350 to blocked.
F6-07	(on rejection)	A return delivery is created (movement 122) and a credit note is expected from the supplier.
F6-08		`quality.usage_decision` → scorecard (OTIF, PPM) and digital-twin feedback. The GR is posted to ERP (exactly-once).

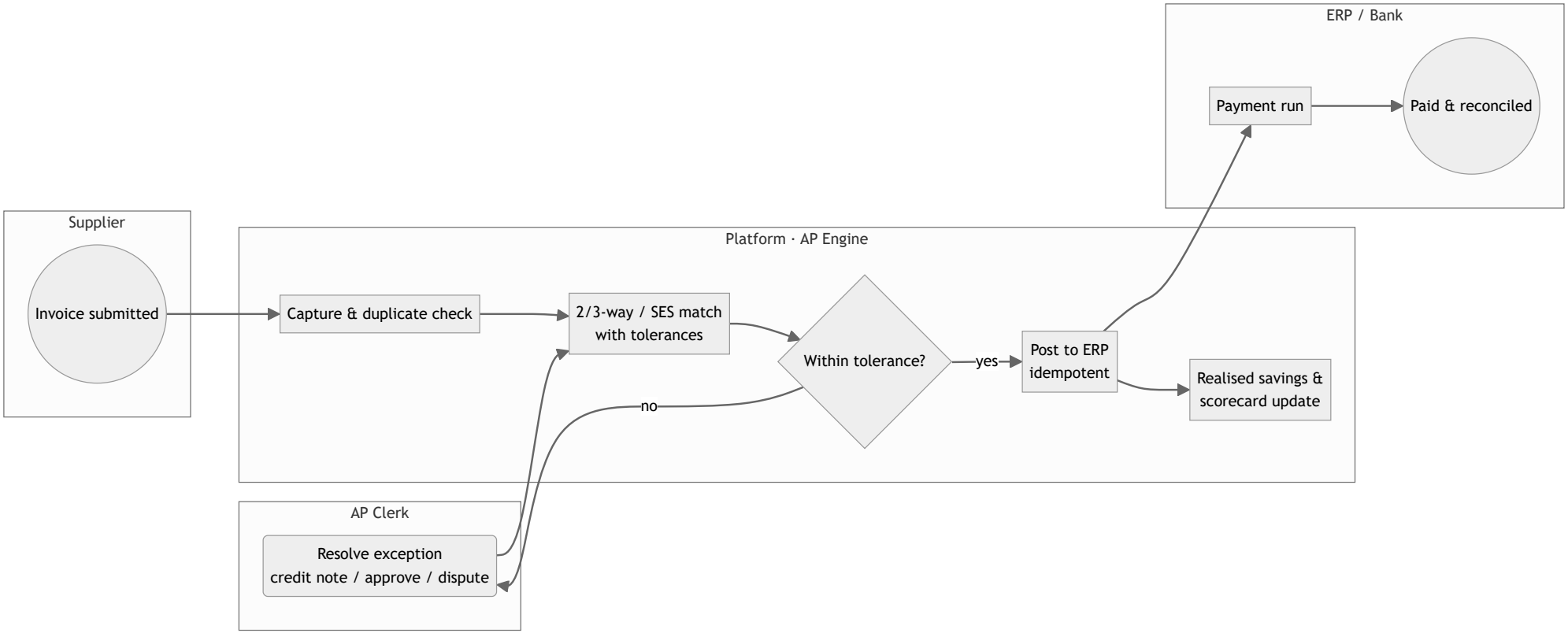
Step	API	What the code enforces
F6-04	`POST /v1/service-entries`, `POST …/approve`	Service owner approval; the supplier user and the PO creator are both excluded. Value ≤ PO line. `service_entry.approved` accrues GRIR.
reversal	`POST /v1/goods-receipts/{id}/reverse`	Not allowed once invoiced and posted (use a return delivery instead).

Recorded paths:

- **GR-90231:**
  - F6-03 5,400 against PO-77410/ASN-61208. F6-05 yes, quality stock.
  - F6-06 QL-2210: 5,380 accepted, 20 rejected (POROSITY).
  - F6-07 RD-4410: 20 returned, credit note requested.
  - F6-08 ERP material document 5000071230. F6-09 partially rejected.
- **SES-3301:** F6-02 service → F6-04 approved by SERVICE_OWNER → F6-08 → F6-09.

F7 — Invoice → match → ERP posting → payment

Process key `pic.invoice` · Module M10 · Service `services/payables.py`



Step	API	What the code enforces
F7-02	`POST /v1/invoices`	Unique key (tenant, company code, supplier, invoice number, fiscal year) → 409 DUPLICATE_INVOICE. Fuzzy duplicate (amount ± ₹1, date ±3 days) → hold. GST e-invoice IRN verified as SHA-256 of GSTIN + FY + type + number.
F7-03	(on capture) / `POST …/match`	**DMN-03** per line: 3-way against accepted GR − already invoiced (0% qty); 2-way against PO; services against approved SES value; price ±1% or ₹500, whichever is lower; tax rate, HSN, tax amount and IRN. Credit notes reduce the matched quantity.
F7-04/05	`POST …/credit-notes`, `POST …/exceptions/{id}/resolve`	Payment block until every exception is resolved. The Invoice Exception agent (L2) explains the cause from GR, inspection and return-delivery evidence. **SOD-04**: the PO creator cannot resolve exceptions on that PO.

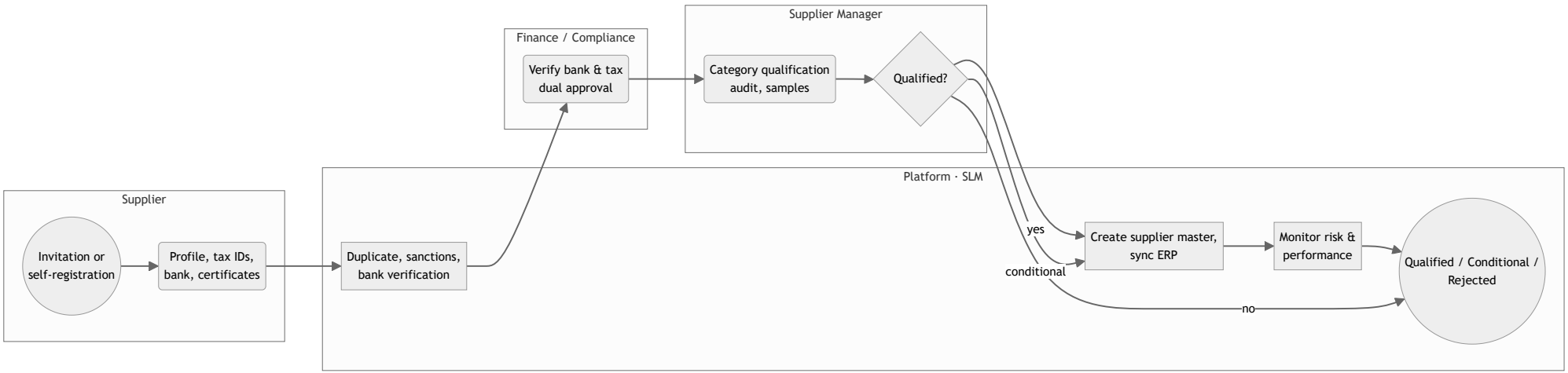
Step	API	What the code enforces
approve	`POST …/approve`	**SOD-02a**: whoever changed the supplier's bank details cannot post its invoices.
F7-06	`POST …/post` with `Idempotency-Key: tenant|company code|supplier|invoice no`	Key format enforced (P13). `integration_message` is unique per key. A retry returns `replayed: true` and the same ERP document. A lost ERP response is resolved by query-before-retry (tested).
F7-08	consumers of `invoice.posted`	Realised savings (only from posted invoices, P12); spend line with maverick flags; scorecard invoice accuracy; budget commitment → consumption.
F7-07/09	`POST /v1/payments/runs`	Pays POSTED, unblocked, due invoices, net of credit notes. Skips suppliers still in the bank-change cooling-off period; alerts when bank details changed within 30 days; **SOD-02**. A partial unique index allows one live payment per invoice, so double payment is impossible (tested).

Recorded path, INV-TF-8812:

Step	Outcome
F7-02	captured; no duplicate; IRN valid
F7-03	3-way: QUANTITY +20
F7-04	no → payment block
F7-05	credit note CN-TF-221 received (ap.clerk)
F7-03	3-way: all lines within tolerance
F7-06	posted as ERP **5100004421** (key AXN|1000|TF001|INV-TF-8812)
F7-08	realised savings ₹64,560.00
F7-07	PRUN-2611: PAY-880002 ₹25,26,663.20 (gross − CN)
F7-09	paid; ERP clearing 1500002211; reconciled

F8 — Supplier onboarding & qualification

Process key `pic.supplier-onboarding` · Module M06 · Service `services/supplier.py`



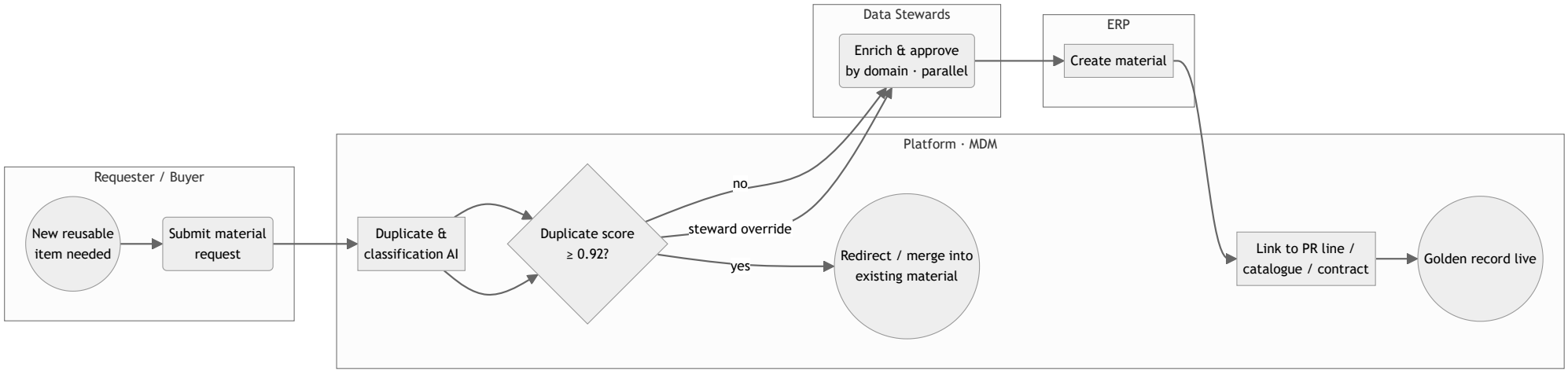
Step	API	What the code enforces
F8-01	`POST /v1/suppliers`	GSTIN encrypted (per-tenant key) with a blind index; PAN hashed. **SOD-01** action logged for the creator. Category status REGISTERED.
F8-02	`POST /v1/suppliers/{code}/profile`	Supplier user or supplier manager. Certificates and the bank account (encrypted, masked) are stored and a bank change request is opened.
F8-03	(on profile)	Duplicate screen (GSTIN blind index, legal-name similarity) and sanctions list; a sanctions hit blocks the supplier.
F8-04	`POST /v1/bank-changes/{no}/callback`, then tasks	The call-back must be done by someone other than the requester. Two **different** finance controllers approve. 72-hour cooling-off before the first payment. **SOD-02** is logged for the requester.
F8-05/06	`POST /v1/suppliers/{code}/qualifications`, `POST /v1/qualifications/{id}/assessment`, `…/decision`	Per category. Mandatory certificates (ALU-DIECAST: IATF 16949, ISO 14001) must be valid and verified; questionnaire ≥ 70, audit ≥ 75; PPAP sample. CONDITIONAL needs a waiver.
F8-07		Status ACTIVE; supplier master posted to ERP (vendor number).
F8-08	timers	Certificate expiry → CONDITIONAL automatically. `POST /v1/suppliers/{code}/block` blocks new POs and opens a buyer task for each open PO.

Recorded paths:

- **TF001:**
  - F8-02 profile with 2 certificates. F8-03 screening CLEAR, then bank call-back verified.
  - F8-04 FINANCE_CONTROLLER → FINANCE_CONTROLLER.
  - F8-05/06 ALU-DIECAST qualified. F8-07 ERP vendor 300001. F8-09 QUALIFIED.
- **MPC005:** waiting at the bank call-back (F8-03), as an example of an in-flight onboarding.

F9 — Material master governance

Process key `pic.material-request` · Module M03 · Service `services/mdm.py`



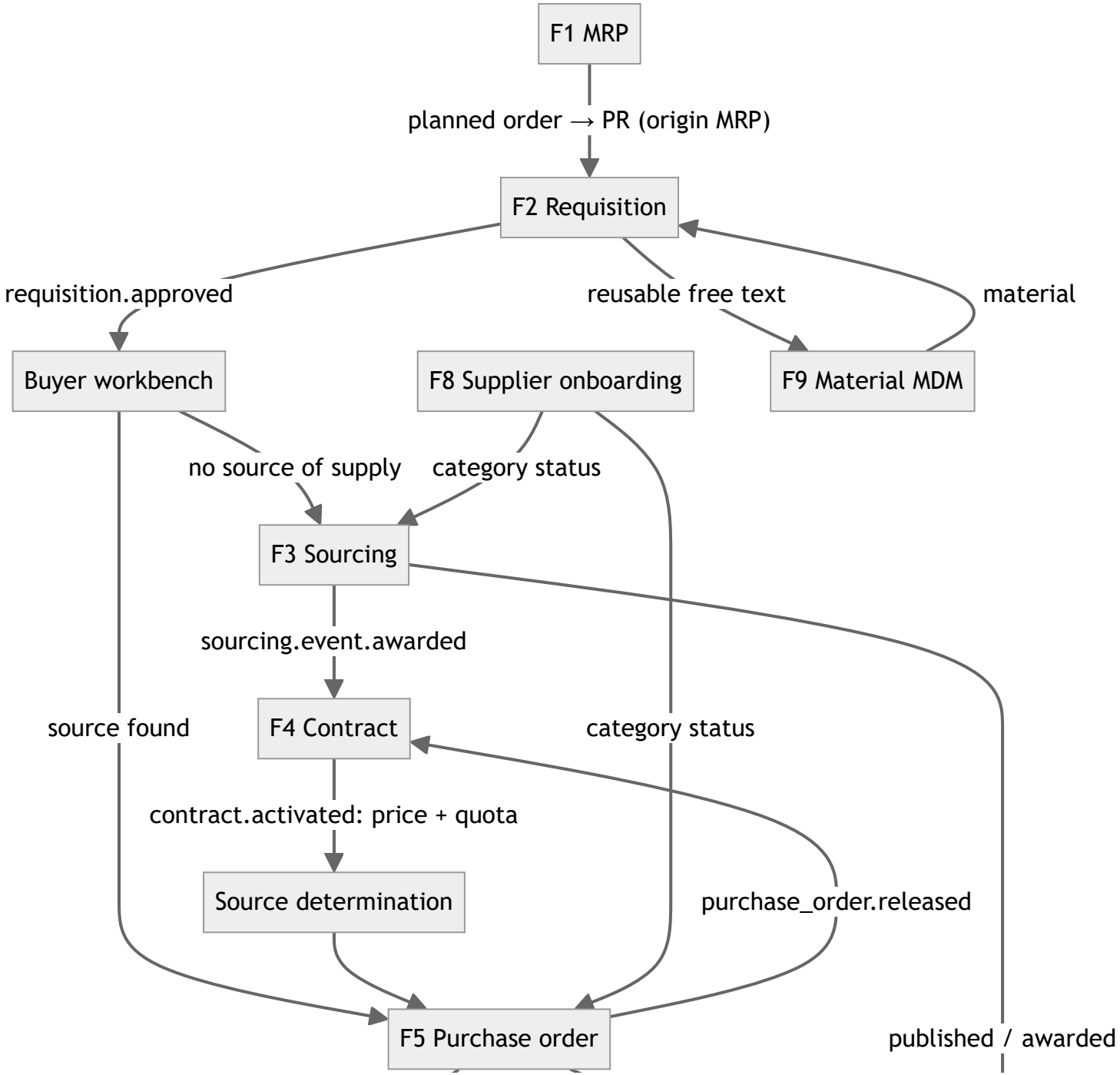
Step	API	What the code enforces
F9-02	`POST /v1/materials/requests` or `POST /v1/requisitions/{id}/lines/{n}/promote-material`	Only free-text lines can be promoted; the PR line stays free text (P04).
F9-03/04	(on submit)	Score = 0.5 description + 0.3 MPN + 0.1 attributes + 0.1 classification. 0.92 or more blocks; 0.75–0.92 warns. Material Dedup agent run (L2).
blocked	`POST …/merge` or `POST …/override`	Merge redirects the requester to the existing material. An override needs a steward and a reason (**SOD-06**).
F9-05	`POST /v1/materials/requests/{id}/decision` or tasks	Four parallel steward tasks (purchasing, quality, finance, plant), each matched by the steward's domain. The requester is excluded (**SOD-06**).
F9-06/07/08		Material created, posted to ERP exactly once, linked to the PR line; `material_request.approved`.

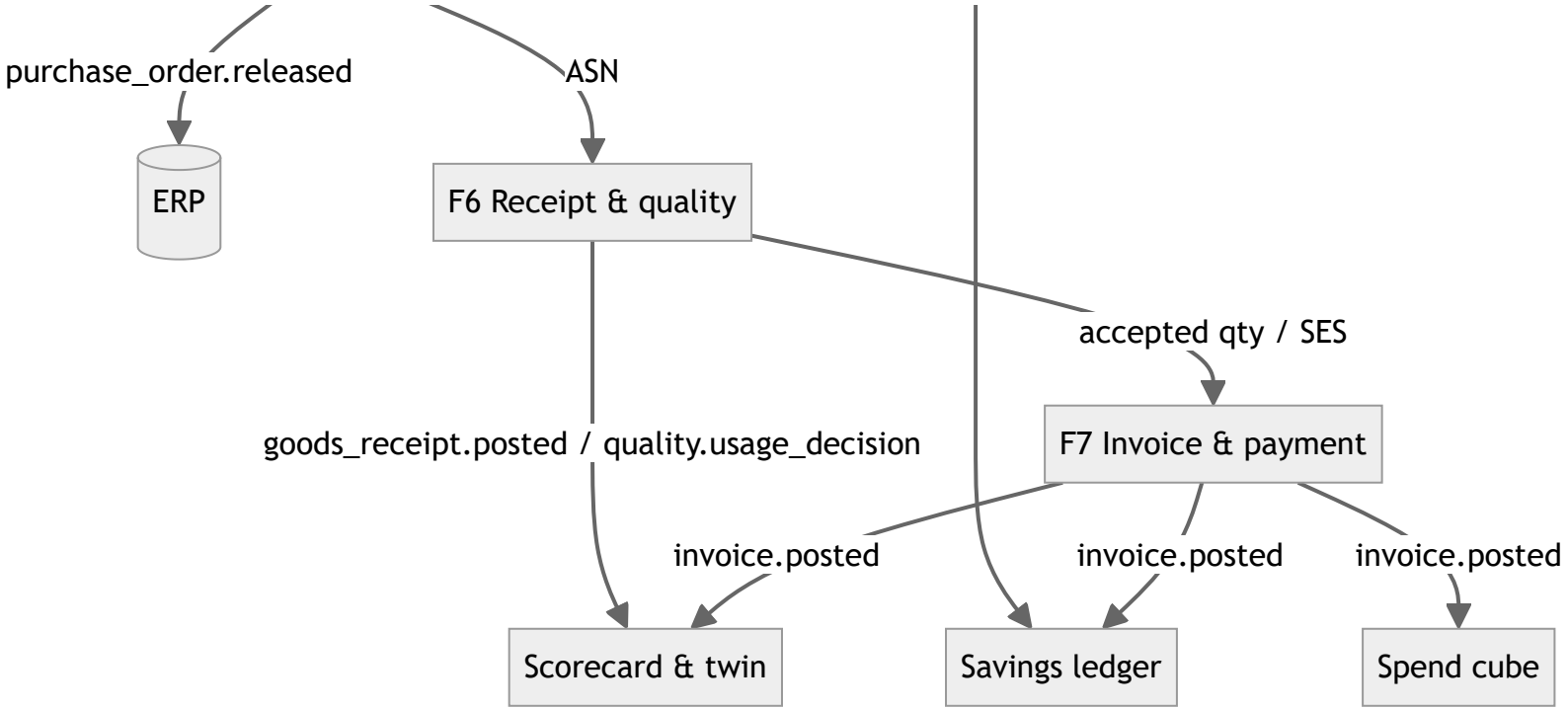
Recorded paths:

- **MRQ-7731:** score 0 → 4 stewards approved → ERP material 100000331 → linked to REQ-2312 line 10 → golden record M-400001.
- **MRQ-7732:** score 1.0 → blocked → buyer redirected and merged into DC-330.

---

10. How the flows connect (events)


F1 MRP
planned order → PR (origin MRP)
F2 Requisition
requisition.approved
reusable free text
material
Buyer workbench
F8 Supplier onboarding
F9 Material MDM
no source of supply
category status
F3 Sourcing
sourcing.event.awarded
source found
F4 Contract
category status
contract.activated: price + quota
Source determination
purchase_order.released
F5 Purchase order
published / awarded

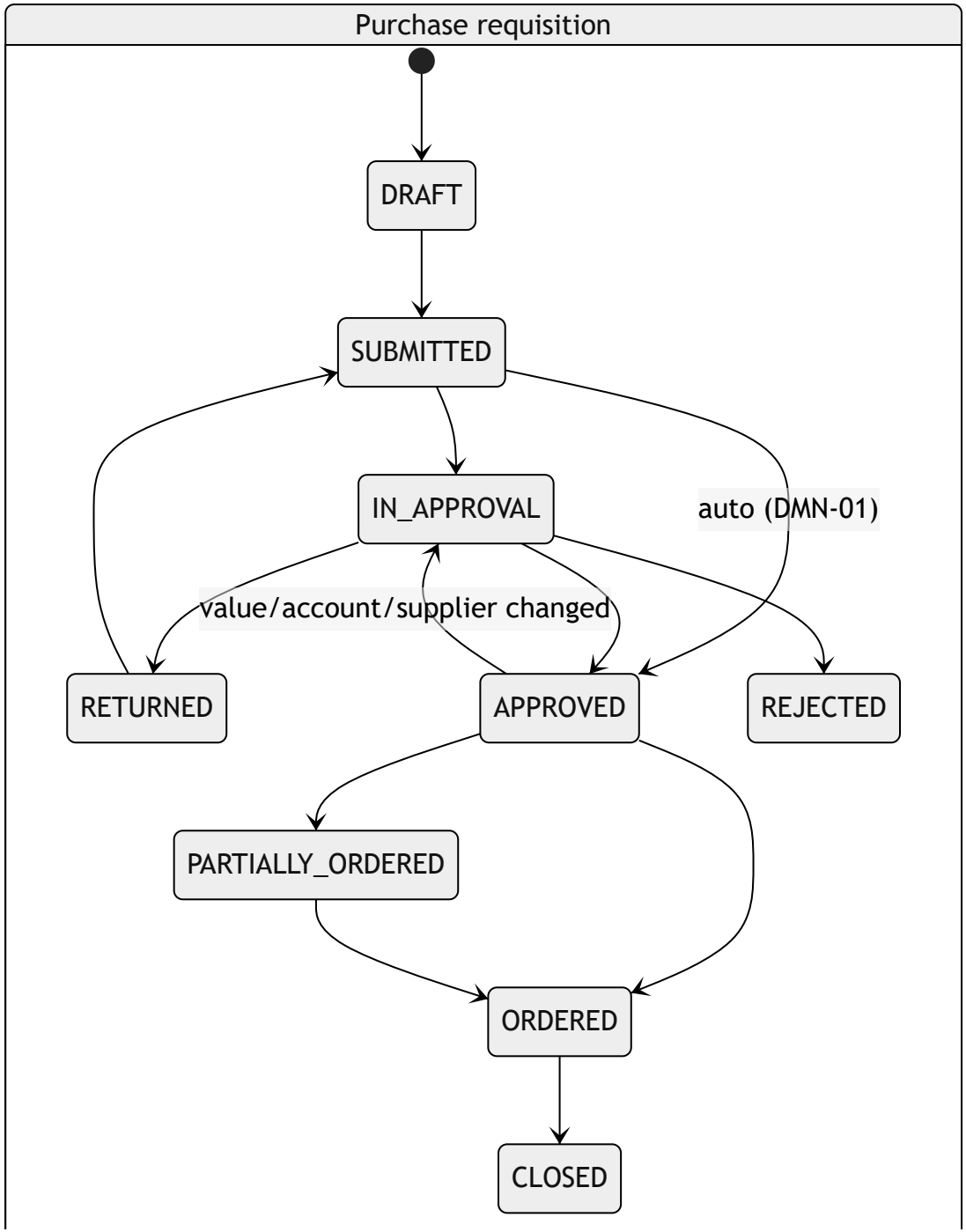


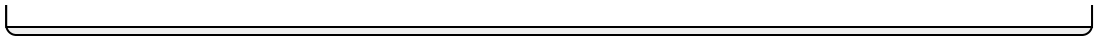
Lineage: every hop writes a `document_link` row. `GET /v1/lineage/{type}/{ref}` returns the graph. PO-77410's chain is PLO-55821 → REQ-2311 → RFQ-2231 → SA-5520, and it continues downstream to ASN-61208, GR-90231, INV-TF-8812, CN-TF-221, ERP 5100004421 and PAY-880002.

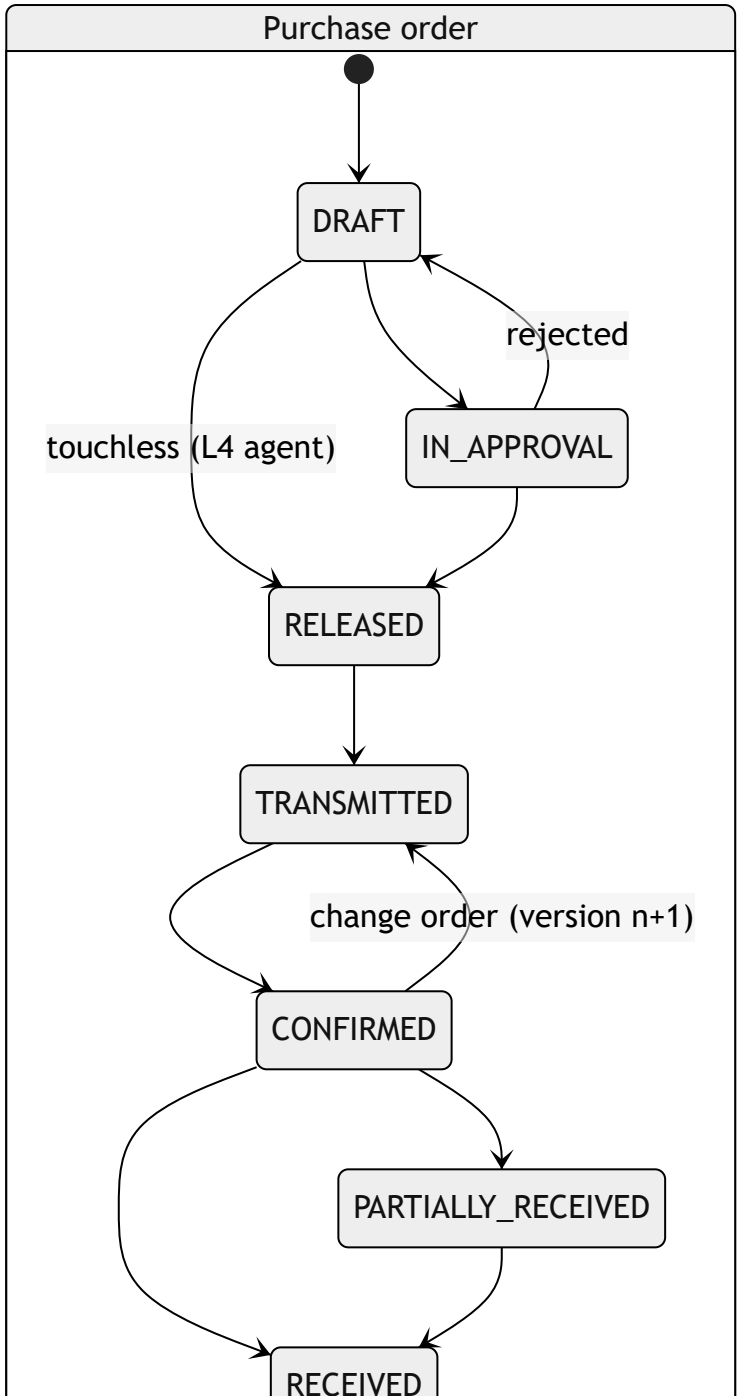
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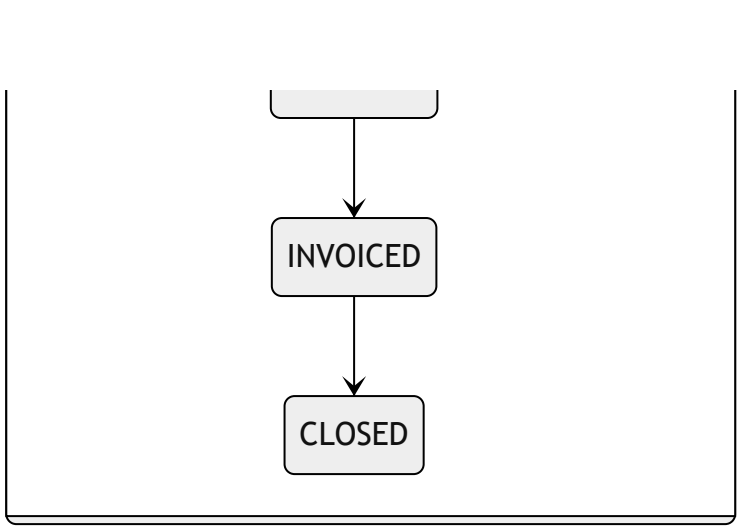
11. Document state machines

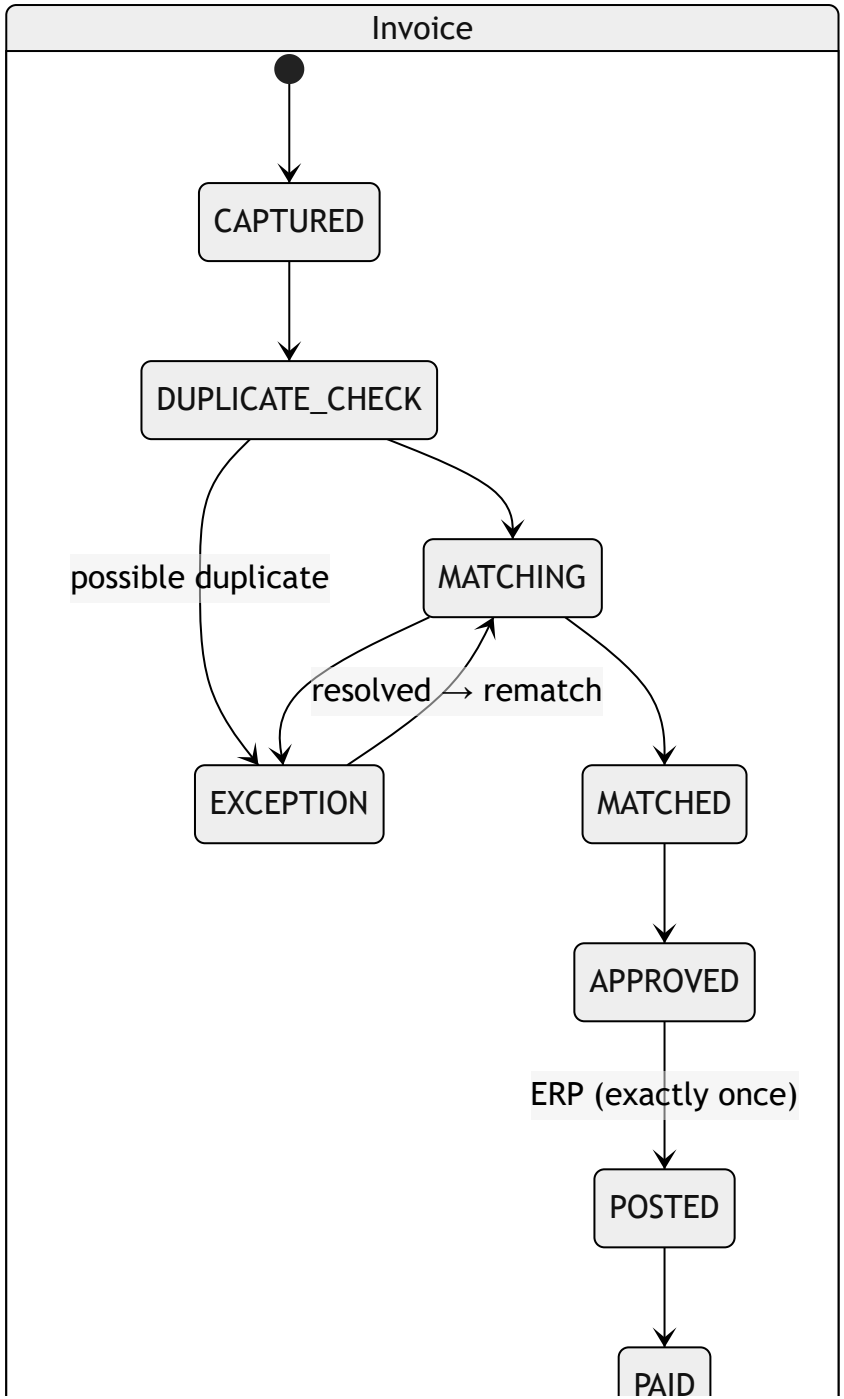
Every command goes through these transitions. Any other transition returns **409 ILLEGAL_TRANSITION**.


Purchase requisition
DRAFT
SUBMITTED
IN_APPROVAL
auto (DMN-01)
value/account/supplier changed
RETURNED
APPROVED
REJECTED
PARTIALLY_ORDERED
ORDERED
CLOSED




Purchase order
DRAFT
rejected
touchless (L4 agent)
IN_APPROVAL
RELEASED
TRANSMITTED
change order (version n+1)
CONFIRMED
PARTIALLY_RECEIVED
RECEIVED


INVOICED
CLOSED


Invoice
CAPTURED
DUPLICATE_CHECK
possible duplicate
MATCHING
resolved → rematch
EXCEPTION
MATCHED
APPROVED
ERP (exactly once)
POSTED
PAID



Object	States (implemented)
Sourcing event	DRAFT → PUBLISHED → OPEN → CLOSED → UNDER_EVALUATION → (OPEN for another round) → AWARD_PENDING → AWARDED · CANCELLED
Contract	DRAFT → IN_NEGOTIATION → IN_APPROVAL → OUT_FOR_SIGNATURE → ACTIVE → EXPIRING → EXPIRED · TERMINATED · RENEWED
Goods receipt	POSTED → QUALITY_HOLD → ACCEPTED · PARTIALLY_REJECTED · REJECTED → REVERSED
Service entry	DRAFT → SUBMITTED → APPROVED · REJECTED → REVOKED
Planned order	CREATED → FIRMED → CONVERTED · DELETED
Material request	DRAFT → DUPLICATE_CHECK → IN_STEWARDSHIP · DUPLICATE_BLOCKED → APPROVED → SYNCED_TO_ERP · REJECTED · MERGED
Supplier (per category)	REGISTERED → UNDER_QUALIFICATION → QUALIFIED · CONDITIONAL → PREFERRED · BLOCKED · EXITED
Savings initiative	IDENTIFIED → NEGOTIATED → CONTRACTED → REALISING → VALIDATED · LOST

---

12. Controls reference

Segregation of duties (checked at command time)

Rule	Conflict	Checked when	Result
SOD-01	supplier-master creator ↔ PO for that supplier	PO create	409 SOD_CONFLICT + `sod_violation` row
SOD-02 / 02a	bank-detail changer ↔ payment approval / invoice posting	payment run, invoice approve/post	blocked / skipped
SOD-03	PO creator ↔ goods receipt on that PO	GR post	409 (tested)
SOD-04	PO creator ↔ invoice exception approval on that PO	exception resolve	409
SOD-05	sole evaluator ↔ award	award proposal and approval	409
SOD-06	material requester ↔ steward approval	steward task, override	409 / excluded
SOD-07	savings owner ↔ validation	savings validate	409

Four-eyes on every approval chain: the initiator and earlier approvers in the chain are excluded from later tasks. The bank change requires two different finance controllers.

Principles P01–P14 → where enforced

ID	Enforcement in code	Proof
P01	submit guard: purpose code	422 RULE_VIOLATION
P02	DB trigger `requisition_origin_immutable`	test_requisition_origin_is_immutable_p02
P03	guided-buying search order	`GET /v1/guided-buying/search`
P04	DB CHECK `free_text_no_material`; promotion keeps the line free text	test_free_text_line_cannot_reference_material_p04
P05	F9 duplicate gateway + parallel stewards	MRQ-7731 / MRQ-7732
P06	event guards: approved requirement, qualified invitees, weights = 100	test_unqualified_supplier_cannot_be_invited, test_weights_must_sum_to_100
P07	PO inherits contract price/terms; deviation reason CHECK	test_po_77410_inherits_contract_and_is_touchless
P08	PURCHASE-TYPE-MATRIX rule set → GR/IR/SES flags	SES flow; service lines refused at GR
P09	workflow tasks + SoD engine	test_sod03…, test_four_eyes…
P10	`document_link` on every hop	lineageChain

ID	Enforcement in code	Proof
P11	`agent_run` with inputs, evidence, model version, autonomy level	`GET /v1/agents/{code}/runs`
P12	ledger: negotiated (award), avoidance (index), realised (posted invoices only)	test_savings_three_kinds
P13	`integration_message` unique key + query-before-retry + replay	test_invoice_post_is_idempotent, test_exactly_once_when_erp_response_is_lost
P14	effective-dated `rule_set`; new versions via `POST /v1/admin/rules`	`rule_set_version` on every document

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13. Signature case: end-to-end results

Blueprint figure	Result in the database
MR-1043 net 9,000	MR-1043 net **9,000** → PLO-55821 (need 12 Nov 2026, release 30 Oct)
REQ-2311 from PLO-55821	REQ-2311, origin MRP, 4-step approval, ORDERED
RFQ-2231 412 / 398 / 405; should-cost 391; baseline 410	Same; should-cost **391.00**; baseline 410 from the info record
Dual source 60/40, weighted 400.8	CAPACITY_CONSTRAINED: TF001 0.60 @ 398, KD002 0.40 @ 405, **400.80**
SA-5520 scheduling agreement, LME quarterly, 60-day terms	SA-5520 (TF001, N60, FCA Trichy, +2/−5) and SA-5521 (KD002) ACTIVE
PO-77410 risk 18, touchless	PO-77410 5,400 @ 398, risk **18**, released by agent:TOUCHLESS_PO; PO-77411 3,600 @ 405
Confirmed in 6 h; ASN with batches and CoC	Confirmed in 6.00 h; ASN-61208, 2 batches, CoC, e-way bill
GR-90231: 5,400 / 5,380 / 20 (porosity)	GR-90231, QL-2210 sample 200; RD-4410 for 20
INV-TF-8812 → +20 → CN-TF-221 → posted 5100004421	Same; paid PAY-880002 ₹25,26,663.20 on 13 Jan 2027
Negotiated 0.99 Cr/yr; avoidance 0.32 Cr	**0.99 Cr**; **0.32 Cr** (0.19 + 0.13 from the two contracts' Q1-27 reset)
Realised YTD 0.38 Cr (portfolio)	0.01 Cr: realised savings come only from posted invoices, and this case has one (₹64,560)

The seed also adds:

- an F2 free-text requisition with an F9 new material (REQ-2312 → MRQ-7731 → M-400001), and a blocked duplicate (MRQ-7732);
- an F6/F7 service path (REQ-2313 → PO-77412 → SES-3301 → HTS/1107, first-pass SES match);
- four suppliers fully onboarded through F8, and one (MPC005) still in progress.

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14. Running it on Neon

1. Create `backend/.env` from `backend/.env.example`:
   - `DATABASE_URL` is the Neon connection string with `sslmode=require`; use the direct host for setup.
   - Generate `PIC_JWT_SECRET` and `PIC_FIELD_KEY`.
   - Optionally set `PIC_SEED_PASSWORD`.
2. From `backend/`, run:

   ```
   venv/Scripts/python -m app.pic.tools.neon_setup
   ```

   This does four things:
   - applies the PIC migrations (schema, role, RLS);
   - seeds the Axion case, skipped if tenant AXN exists;
   - loads the legacy data from `backend/data/legacy/supplier_twin.json` (see §17): it creates the legacy tables, copies all 15 legacy tables verbatim, migrates them into PIC tenants GMC and DBY, and writes `docs/PIC_Legacy_Migration_Report.md`;
   - prints a verification report covering row counts, RLS coverage, audit chains and pending events.
3. Run the API with `uvicorn app.main:app --port 8080`. Interactive docs are at `/docs`; the PIC routes are under `/v1`.
4. Run the worker with `python -m app.pic.worker`. It relays the outbox, fires timers and retries ERP postings.
5. Log in with `POST /v1/auth/token {"tenant":"AXN","email":"buyer@axion.example","password":"…"}`.
6. Run the tests with `python -m pytest tests/pic`. They need `TEST_DATABASE_URL`, a disposable database; they refuse to run against `neon.tech`.

Demo users (tenant AXN) — all `…@axion.example`:

Area	Users
Admin and planning	`admin`, `mrp.controller`, `requester`
Approvers	`plant.controller`, `ops.head`, `procurement.head`, `cpo`, `finance.controller`, `finance.controller2`, `legal`
Buying, sourcing, contracts	`buyer`, `buyer.lead`, `category.manager`, `category.head`, `contract.manager`, `supplier.manager`
Data stewards	`mdm.purchasing`, `mdm.quality`, `mdm.finance`, `mdm.plant`
Receiving, quality, service, AP	`receiver`, `quality.engineer`, `quality.inspector`, `service.owner`, `ap.clerk`
Supplier users	`tf.sales`, `kd.sales`, `sla.sales`, `hts.service`, `mpc.sales`

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15. Operating cases (seeded after the signature case)

These cases run through the same services as the signature case (`backend/app/pic/seed/operations.py`, Jan–Mar 2027). Each one exercises a branch that the main path does not.

#	Case	Flow branch exercised	Result in the data
1	February DC-330 release, 9,000 pcs	F2 manual stock PR → F5 quota split → F6 short shipment → F7 price exception	REQ-2315 → PO-77414 (TF 5,400) and PO-77415 (KD 3,600), both touchless. KD ships 3,420 (−5%, inside tolerance), so the receipt completes. INV-TF-9031 matches first time. KD/2702-118 bills 410 against the 405 contract → price exception → variance accepted with a note → **savings leakage** of ₹17,100 on SV-0412.
2	MRO catalogue buy, ₹18,500	F2 auto-approval (catalogue < ₹25,000) → F5 touchless catalogue PO → F6 service entry	REQ-2314 approved by DMN-01 with no human → PO-77413 released by the L4 agent → unit-rate SES → HTS/1188 SES match → posted.
3	Duplicate invoice	F7 fuzzy duplicate hold → reject	KD/2702-118-R (same amount, date within 3 days) is held as an exception and rejected; nothing posted or paid.
4	Conditional supplier	F8 → CONDITIONAL with waiver	MPC005: bank call-back and dual approval, then qualified CONDITIONAL with the ISO 14001 waiver recorded.
5	Unqualified supplier + SoD	F5 risk +40 → approval; F6 SOD-03 block	PO-77416 (fasteners from KD002, not qualified for FASTENERS) is risk-scored 40 and approved by the buyer lead. The stores buyer who created it is blocked from posting its GR (SOD-03, logged in `sod_violation`); the receiver posts it.
6	Sanctions screening	F8-03 screening → automatic block	ZAT006 "Zenith Alloy Trading FZE" fuzzy-matches the screening-list entry "Zenith Alloys Trading FZE" and is **BLOCKED** at registration. Until the licensed feed is connected, the list (`sanction_entry`) holds fictitious `SAMPLE-LIST` names.
7	Source list → change order	F5 source determination (source list + info record) → approval → confirmation → change order → re-confirmation	REQ-2317 (DC-331, 2,000) → PO-77417 from the ERP source list: Trichy Forge is the fixed source, at the info-record price ₹212. The PO is not touchless (the price source is not a contract or catalogue), so it is approved and confirmed. The OEM then pulls the schedule in: the buyer raises the quantity to 2,400 and moves the date to 18 Mar. This creates **version 2** (`po_version`), the PO goes back to TRANSMITTED, and Trichy Forge re-confirms.
8	Supplier change request after confirmation	PO-03: supplier proposes → buyer task `PO_CHANGE_REQUEST` → change order → re-confirmation	Trichy Forge raises **PCR-501** on PO-77417 (furnace relining on cell 2, ship 20 Mar instead of 18 Mar). The buyer lead accepts it, which creates **version 3**; Trichy Forge re-confirms the same afternoon.
–	Month-end payment run, 31 Mar 2027	F7-07	PRUN-2612 pays HTS/1188 and KD/2702-118. INV-TF-9031 (N60, due 14 Apr) stays open, as expected.

16. Workspace screens (web)

The web workspace is at `/pic`. Sign-in is at `/pic/login`; supplier users are sent to `/pic/portal`. Every screen reads live data through the web server's `/api/pic` route, which attaches the httpOnly session token.

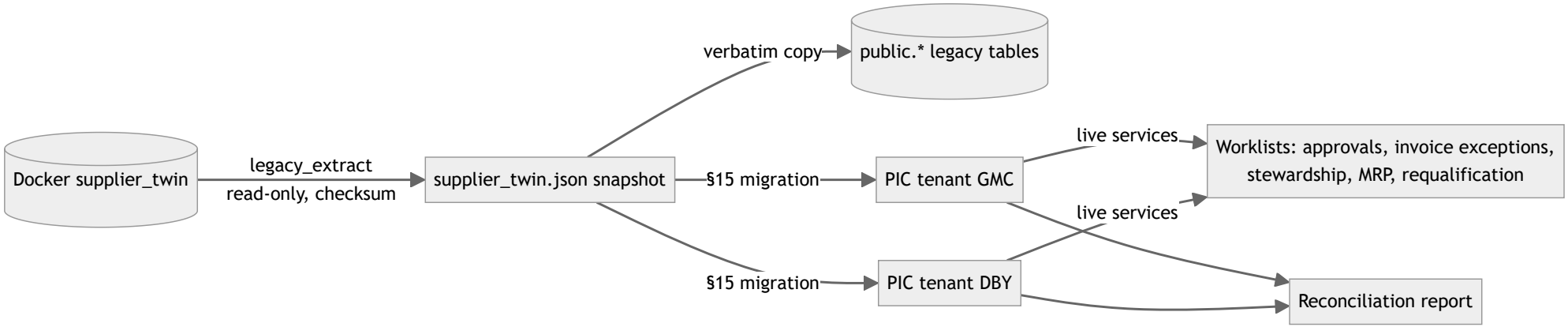
Area	Route	Blueprint screens
Command centre	`/pic`	AIX-01 (live KPIs, live event stream, recent audit)
Task inbox	`/pic/tasks`	REQ-03 (approve / return / reject with comment, four-eyes)
MRP	`/pic/mrp`, `/pic/mrp/[run]`	MRP-01/02/03 (run console, netting trace, firm / convert)
Requisitions	`/pic/requisitions`, `/new`, `/[ref]`	REQ-01/02 (guided search in enforced order, policy and budget feedback)
Buyer workbench	`/pic/buyer`	REQ-04 (source determination, create PO, risk and lineage result)
Materials	`/pic/materials`, `/[ref]`	MDM-01/02/03 (duplicate score, stewards, merge / override)
Sourcing	`/pic/sourcing`, `/new`, `/[ref]`	SRC-01..05 (builder, sealed bids, TCO matrix, scenarios, award)
Contracts	`/pic/contracts`, `/[ref]`	CLM-01..04 (review, approval, signatures, activation, index reset, consumption)
Suppliers	`/pic/suppliers`, `/new`, `/[code]`	SLM-01/02 (360 view, onboarding, bank call-back, qualification, block)
Purchase orders	`/pic/purchase-orders`, `/[ref]`	PO-01..04 (validation explained, change order, ASN)
Receiving	`/pic/receiving`, `/[ref]`	RCV-01..04 (inbound, GR, usage decision, SES, reversal)
Invoices	`/pic/invoices`, `/[id]`	INV-01..04 (capture, match detail, exceptions, credit notes, post, payment run, GR/IR)
Savings, spend	`/pic/savings`, `/pic/spend`	SAV-01/02 (+ new initiative), SPD-01..09 (overview, categories, suppliers, cube, compliance, maverick, classification review, category strategy, forecast)
Demand forecast	`/pic/mrp/forecast`	MRP-04 (bias, MAPE, override with reason, release to MRP)
Supplier performance, risk	`/pic/suppliers/performance`, `/pic/suppliers/risk`	SLM-03 (scorecards, CAPA, preferred / exit), SLM-04 (risk heat map, signals, mitigation)
Obligations, new contract	`/pic/contracts/obligations`, `/pic/contracts/new`	CLM-04, CLM-03 renewals calendar, CLM-01 manual contract
PO changes, expediting	`/pic/purchase-orders/changes`, `/expediting`	PO-03, PO-04
Reconciliation, insights	`/pic/invoices/reconciliation`, `/insights`	INV-03, INV-04

Area	Route	Blueprint screens
Copilot	`/pic/copilot`	AIX-02 (answer contract rendered in full)
Process monitor	`/pic/processes`	F1–F9 swimlanes and live instances
AI agents	`/pic/agents`	AIX-03 (autonomy editor, run now, recommendations inbox, runs, kill switch)
Administration	`/pic/admin`	ADM-01/02/04/05/06 (organisation, currencies and FX, users and roles, industry templates, rule versions, SoD, integration replay, audit + hash-chain check)
Supplier portal	`/pic/portal`	SLM-05 (confirm, ASN, service sheet, sealed bid, invoice, credit note, sign / redline)

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17. Legacy data migration (Docker supplier_twin)

The legacy app's live data was in the Docker Desktop container `supplier_twin_db` (PostgreSQL 15, database `supplier_twin`, 15 tables, 156 rows). The other containers on the machine belong to other projects and were not touched. The full reconciliation and the findings are in **`docs/PIC_Legacy_Migration_Report.md`**.



Step	Command / module	What it does
Extract	`python -m app.pic.tools.legacy_extract`	Runs `psql` inside the container (no password handled) and writes `backend/data/legacy/supplier_twin.json` with row counts and a SHA-256. `--url` or `--sqlite` read other sources.
Schema drift	legacy Alembic `e3c1f2a4b5d6`	Adds 9 columns that the models and the Docker DB have but the legacy chain lacked (inventory category/unit/thresholds/location/status; stock-transaction notes/performed-by/balance). Uses `ADD COLUMN IF NOT EXISTS`, so it is safe on both.
Verbatim copy	`neon_setup` step 3a	All 15 tables go into `public` with their ids, `ON CONFLICT DO NOTHING`.
PIC migration	`app/pic/seed/legacy.py`	One PIC tenant per legacy buyer. Masters, open documents and paid history are loaded with legacy ids, audited `MIGRATED` and linked for lineage. The open items then run through the live services.
Report	`app/pic/tools/migration_report.py`	Counts and values legacy vs PIC, the live results and the control findings.

Results (rehearsed on a local PostgreSQL 16 copy of the Neon setup):

Tenant	Loaded	Live continuation	Key findings on the legacy data
GMC: Global Manufacturing Corp	5 suppliers, 5 requisitions, 5 POs, 5 invoices, 2 payments; all counts and values match	PO-2026-6488 validated (risk 58: price deviation, supplier not qualified) → **IN_APPROVAL** task; 3 pending invoices → **MISSING_GR** exceptions; 3 flagged suppliers → requalification	Invoice paid **before the PO was approved**; 2 invoices paid without a goods receipt; PO value ≠ approved quotation
DBY: Dummy Buyer	1 supplier, 9 materials + stock + 13 movements, 6 requisitions, sourcing event RFQ-2026-092 (2 rounds, 280 → 265), 6 POs, 3 ASNs, 3 GRs, 24 invoices, 13 payments; all counts and values match	11 pending invoices → **DUPLICATE** hold; nozzles bought twice as free text → **MRQ-7731** with 4 steward tasks; MRP **MR-1043** on consumption forecast → PLO → **REQ-2312 in approval**; 2 below-safety alerts	**3 overpaid POs** (e.g. PO-2025-00089: 7 invoices paid, 192,000 against 29,500); a disputed invoice paid; 2 POs closed without a receipt; test postings flagged

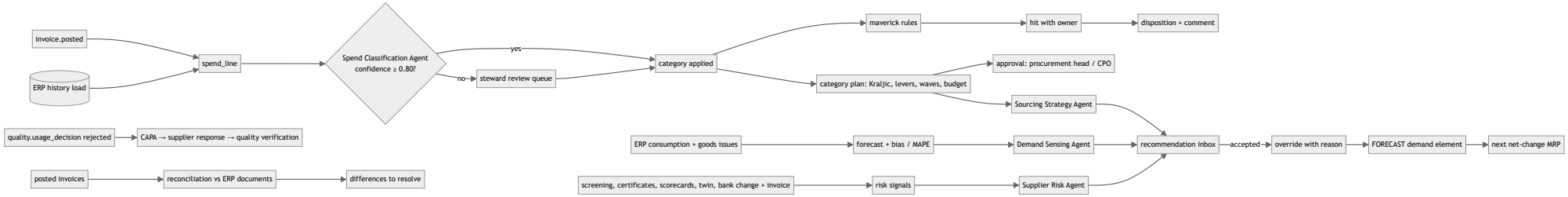
Users of the migrated tenants share the seed password. Internal users are `<role>@gmc.example` or `<role>@dby.example`. Where the legacy logs name a person, that person fills the matching role: for example, Elena Rostova is the DBY finance controller and Marcus Vance the DBY receiver. Legacy users keep their e-mail addresses; for example, `carriedavis@example.net` signs in to GMC's supplier portal as James-Hampton.

After the migration, **every PIC table holds data**. The one exception is `idempotency_record`, which fills at runtime from API calls that carry an Idempotency-Key.

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18. Intelligence and governance modules

These modules sit on top of F1–F9. They read the documents the flows produce, raise work for people and agents, and feed decisions back into the flows. The gap analysis against the blueprint is in `docs/PIC_Module_Gap_Analysis.md`.



Module	Trigger	Human decision	Rule set
Spend classification	Every spend line (on invoice posting or bulk load)	Steward accepts or reclassifies below 0.80	`SPEND-GOVERNANCE`
Maverick rules	Every spend line	Owner disposes: accepted exception / recovery / policy breach / false positive, with comment	`maverick_rule` per tenant
Category strategy	`POST /v1/category-plans`	Procurement head or CPO approves; the owner cannot	—
Forecast review	`POST /v1/forecasts/generate` + Demand Sensing	MRP controller overrides (reason required) and releases to MRP	`DEMAND-SENSING`
Supplier risk	`POST /v1/supplier-risk/scan` or agent run	Supplier manager mitigates / closes	`SUPPLIER-RISK`
CAPA	Quality rejection on a usage decision	Supplier responds; quality verifies effectiveness	—
Preferred / exit	Supplier or category manager	Preferred needs 3-period OTIF ≥ 95% and PPM ≤ 1,000	`SUPPLIER-RISK`
PO change request	Supplier (portal) or buyer	Buyer accepts → version n+1 → re-confirmation	`PO-APPROVAL`
Reconciliation	AP or finance runs it (nightly in production)	Resolve each difference	—
Agents	Events or `POST /v1/agents/{code}/runs`	Accept / reject (reject needs a comment); feedback drives acceptance rate and automatic suspension	autonomy policy per agent

Results on the rehearsal database:

- **AXN, spend:** trailing 12 months (Oct 2025 – Sep 2026) is 415 Cr, 99% of it addressable; 70.5% is contracted and 6.2% maverick. The blueprint's AIX-01 figures are 412 Cr, 71% and 6.2%.

- **AXN, spend governance:** 433 maverick hits, 48 classification reviews and 15 category plans. ALU-DIECAST is approved as strategic.
- **AXN, forecasting:** DC-331 was over-forecast by 22%. The Demand Sensing recommendation was accepted, which set March 2027 to 17,227 and released it to MRP.
- **AXN, suppliers and finance:** CAPA-301 holds Trichy Forge's response, and reconciliation REC-101 found 0 differences.